Solution Requirements

Date	30 June 2026
Team ID	******
Project Name	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The following requirements define exactly what the Smart Lender solution will do and how well it will perform.

Functional Requirements (FR) define the features and operations that the Smart Lender system must perform to predict loan eligibility accurately.

Non-Functional Requirements (NFR) specify the quality standards that ensure the Smart Lender system performs efficiently, securely, and reliably.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium /Low)
1	User Input	The system shall allow users to enter applicant details such as gender, marital status, education, income, loan amount, loan term, and credit history.	High
2	Data Validation	The system shall validate all input fields before processing the application to ensure data accuracy and completeness.	High
3	Loan Eligibility Prediction	The system shall predict loan eligibility using the trained Random Forest machine learning model.	High
4	Prediction Result	The system shall display whether the loan is Approved or Rejected instantly after prediction.	High
5	Model Integration	The Flask application shall load the trained machine learning model (model.pkl) during application startup.	High

S. No	Requirement Category	Requirement Description	Priority (High/Medium /Low)
		risk applicants, exportable as CSV/PDF.	
6	Data Storage	The application shall temporarily process applicant information required for prediction without modifying the trained model.	Medium
7	Error Handling	The system shall display meaningful error messages when invalid or incomplete data is entered.	Medium
8	User Interface	The application shall provide a simple, responsive, and user-friendly web interface for loan prediction.	High

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate loan eligibility predictions quickly after the user submits the application.	Prediction response time < 3 seconds
2	Scalability	The system should support multiple users submitting loan applications simultaneously without affecting performance.	Supports 100+ concurrent requests
3	Security & Data Privacy	Applicant information should be securely processed and protected from unauthorized access.	Secure data handling with server-side validation
4	Reliability & Availability	The application should remain available and provide consistent prediction results with minimal downtime.	99% system availability
5	Usability & Accessibility	The web interface should be simple, responsive, and easy for users to understand and operate.	User-friendly interface with clear validation messages
6	Maintainability	The trained machine learning model should be easy to update or replace without changing the application structure.	New model can be integrated within 1 day